

Lorenz curves: AI Usage Index (AUI) concentration

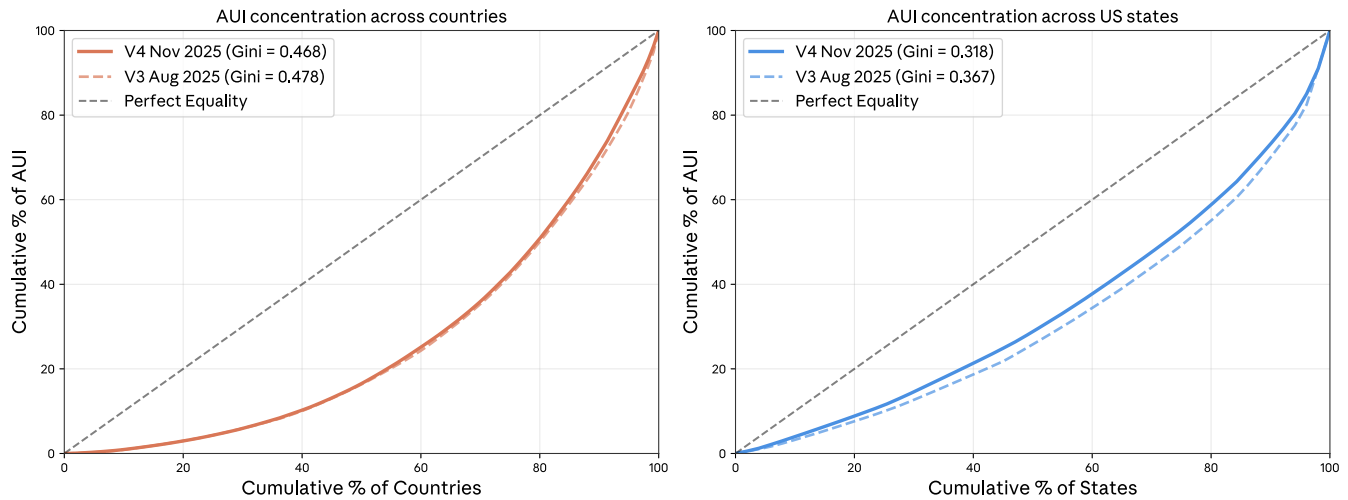


Figure 1.6: AUI concentration around the world and within the US in this and the prior report

Lorenz curves for the Anthropic AI Usage Index (AUI) around the world and within the US, August and November 2025. A curve that is closer to the 45-degree line indicates less concentration. The plot on the right shows, for example, that the top 20 percent of US states accounted for 40 percent of population-adjusted usage in the US.

What shapes patterns of usage within the US and around the world? In our previous report we emphasized the key role played by income differences globally: Variation in Claude usage across countries is largely accounted for by variation in GDP per capita. In Chapter 3 we revisit the importance of income in shaping not just usage intensity but also patterns of usage around the world.

Within the US, income is less clearly a predictor of usage. Instead, what appears to matter most is the composition of each state's workforce and how well-matched the workforce is to Claude capabilities as reflected in task-level usage. States that have a higher share of workers in computer and mathematical occupations—like Washington D.C., Virginia, and Washington—tend to have higher usage per capita. Quantitatively, each 1% increase in the share of such tech workers in a state is associated with 0.36% higher usage per capita (Figure 1.7). This alone accounts for nearly two-thirds of the cross-state variation in AUI.

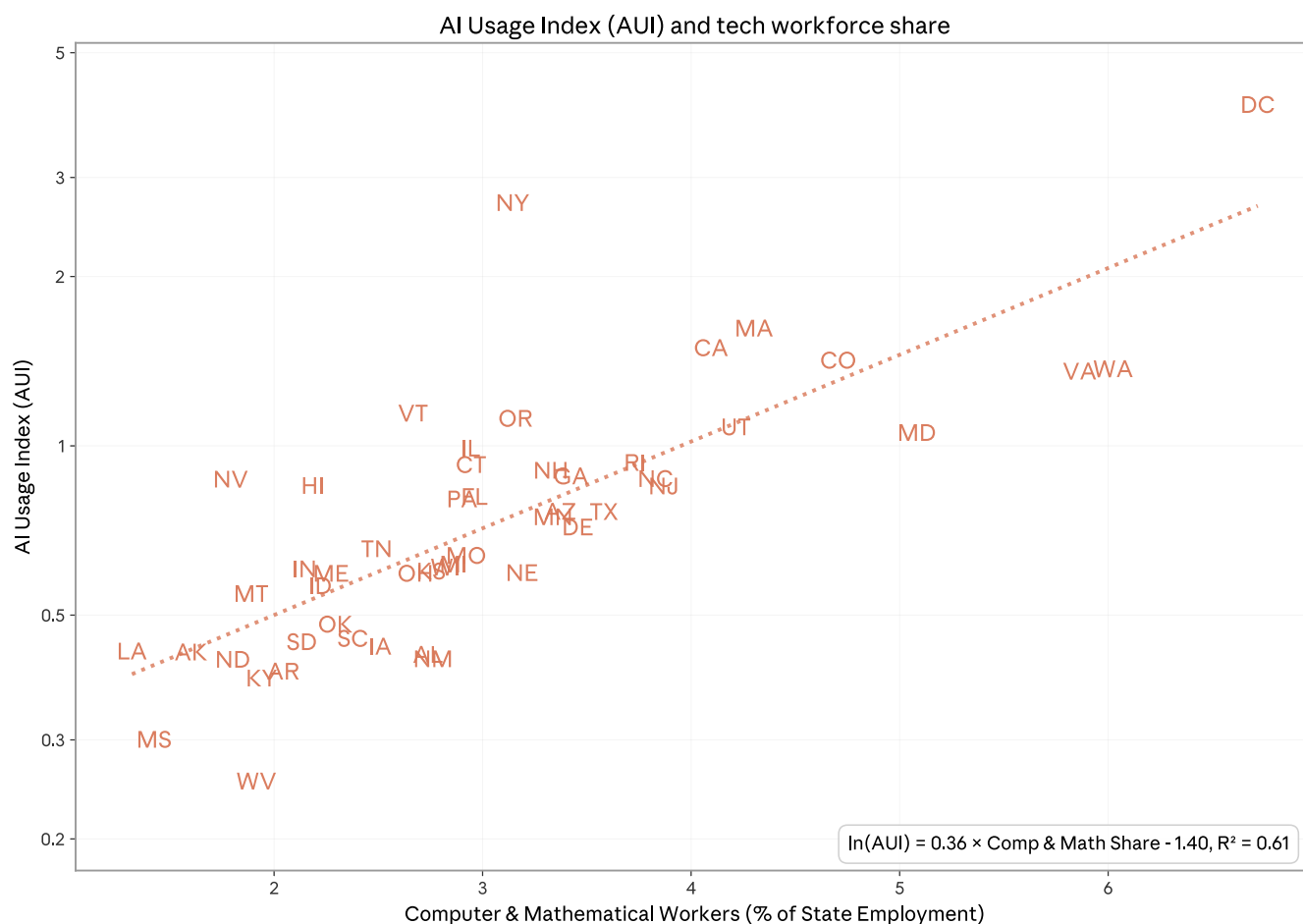


Figure 1.7: AUI and share of workers in Computer & Mathematical occupations in each US State

This figure shows that the share of workers in Computer & Mathematical occupations across US states is highly correlated with the Anthropic AI Usage Index (AUI). This is consistent with the view that overall Claude usage patterns—and associated capabilities—are shaping regional adoption patterns within the US. This pattern holds more generally when formally calculating the KL divergence between each state's workforce distribution and global Claude.ai usage shares by SOC major group.

While we would intuitively expect Claude usage to be higher in states with more tech workers, this pattern holds more generally: Usage per capita is higher in states with more workers in occupations where Claude usage is overrepresented as compared to the US workforce (e.g., Arts, Design, Entertainment, Sports and Media) or with relatively fewer workers in occupations where Claude usage is low as compared to the national economy (e.g., Transportation and Material Moving). This can be seen by calculating the Kullback-Leibler (KL) divergence between the composition of each state's workforce and the global composition of Claude usage. States with a lower KL divergence—and thus with a workforce that looks more similar to Claude usage patterns—tend to have higher usage per capita.

Signs of faster Claude diffusion in the US among low usage states

While differences in workforce composition appear to play a role in shaping regional adoption within the US, early evidence suggests Claude is diffusing considerably faster than historical precedent would predict. Economically consequential technologies have historically taken around half a century to achieve full diffusion across the US ([Kalanyi et al., 2025](#)). By contrast, comparing Claude adoption rates in November 2025 to three months prior, we estimate that parity in adoption per capita across US states—as measured by the AUI—could be reached within 2–5 years. This estimate comes with a high degree of uncertainty as the precision of our estimates cannot rule out much slower rates of diffusion.

We generate this estimate through the lens of a simple model of diffusion, which we briefly describe here. We model diffusion as proportional convergence toward a common steady state of equalized usage per capita in which each state s has an AUI equal to 1:

$$AUI_{s,t} = AUI_{s,t-1}^{\beta}, \beta \in (0,1)$$

Under this model, the log deviation of AUI from steady state ($AUI = 1$) shrinks by a factor of every three months, implying a half-life of $\ln(.5)/\ln(\beta)$ quarters. For example, with quarterly data a value of $\beta = 0.99$ implies a half-life of about 17 years. To illustrate, starting from an initial AUI of 2, this means AUI would decline to around 1.4 after 17 years and to around 1.1 after 50 years. We take $\beta = 0.99$ as a sensible benchmark because it implies a pace of diffusion similar to economically consequential technologies in the 20th century.

This model of convergence motivates the following regression specification:⁸

$$\ln AUI_{s,t} = \alpha + \beta \times \ln AUI_{s,t-1} + \epsilon_{s,t}$$

Naively estimating this equation by ordinary least squares (OLS) yields an estimate of $\hat{\beta} \approx 0.77$. Weighted least squares (WLS) where we weight by each state's workforce yields an estimate of $\hat{\beta} \approx 0.76$ (Figure 1.8). Both are statistically distinguishable from 1 at conventional levels. Taken at face value, these estimates imply that it would take little more than two years for each state's AUI to close most of the gap to 1.

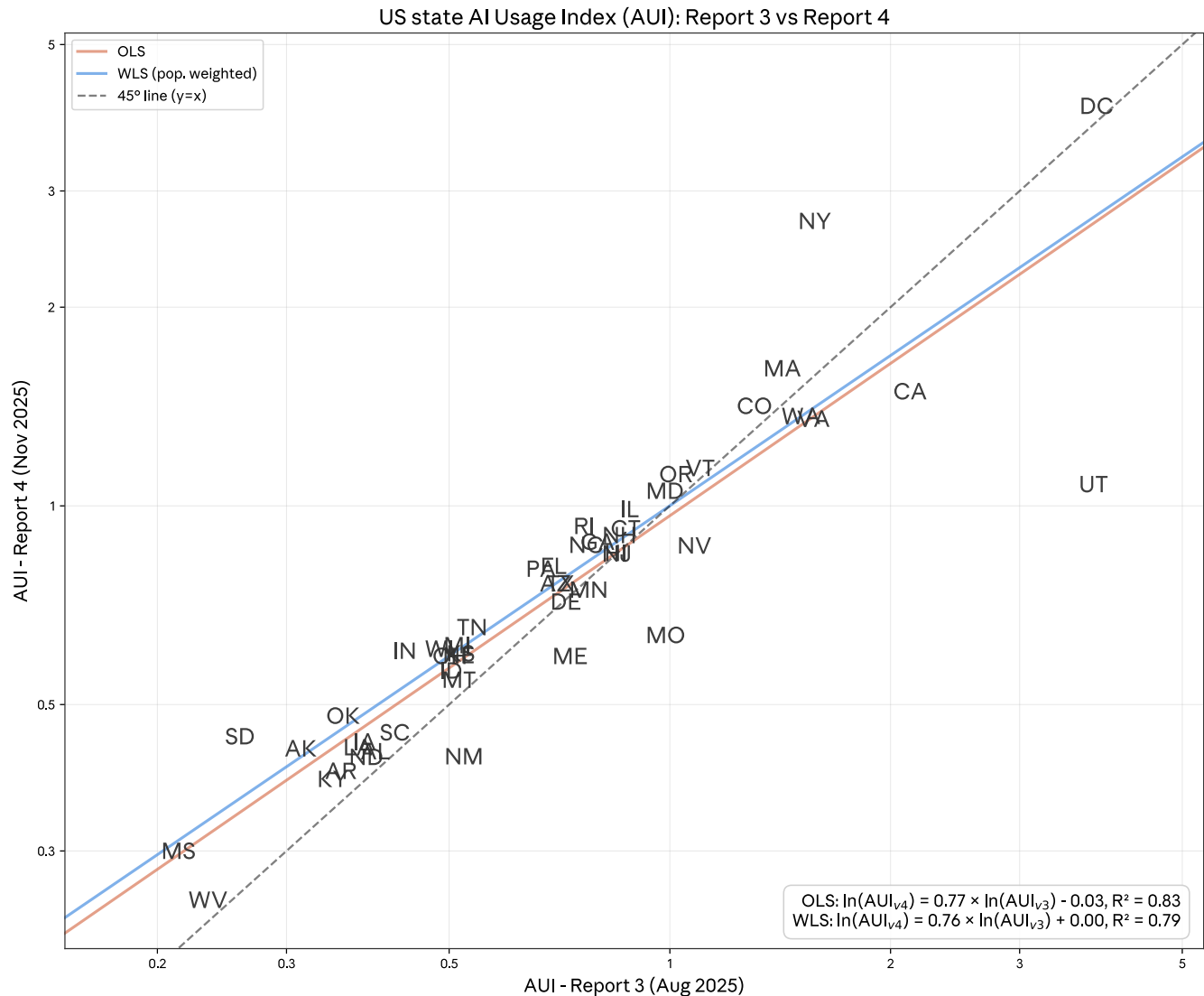


Figure 1.8: Anthropic AI Usage Index (AUI) across the US, August 2025 (V3) and November 2025 (V4)
 By comparing the AUI in November 2025 with its value in August 2025 we can estimate the implied rate of diffusion of Claude usage within the US. Under a model of proportional convergence toward a steady state in which AUI = 1 for all US states, the estimated elasticity can be used to calculate the pace of diffusion (see text for more details). Our range of estimates implies a pace of regional convergence of AUI in 2-5 years.

A concern with estimating convergence this way is that our AUI estimates are subject to sampling noise and other variation unrelated to diffusion. This can produce classical attenuation bias: even if AUI is not actually changing, our estimate of could end up meaningfully below one.

To address this, we estimate the model by two-stage least squares (2SLS), instrumenting the log of AUI in August 2025 with the composition of each state’s workforce, measured by its proximity to overall Claude usage patterns. The logic behind this instrument is that workforce composition is a strong